

FRTT350F Rear Helper Spring Kit Installation Manual

Model: FRTT350F (Rear Suspension Package)

Please read these instructions carefully before starting the installation. Ensure all standard safety precautions are observed while working underneath the vehicle.

1. Parts List

Before beginning, verify that all components listed below are present in your kit.

Ref #	Description / Specifications	Qty
1	Hex Cap Screw (HCS) (3/8" x 1" UNF, Grade 5, Zinc Plated)	2
2	Split Lock Washer (3/8", Grade 5, Zinc Plated)	4
3	Flat Washer USS (3/8" x 1", Grade 2, Zinc Plated)	2
4	Weldment Bracket (Left-Hand & Right-Hand)	2
5	Rubber Spacer with Bushing (4" O.D. x 1" Thick)	2
6	Hollow Rubber Spring	2
7	Hex Cap Screw (HCS) (3/8" x 1-3/4" UNF, Grade 5, Zinc Plated)	2
8	Hex Cap Screw (HCS) (3/8" x 2-3/4" UNF, Grade 5, Zinc Plated)	2
9	Nylock Locknut (3/8" UNF, Grade 5, Zinc Plated)	2
10	Fender Washer (3/8" x 1-1/2", Grade 2, Zinc Plated)	2

2. Installation Instructions

- 1. Raise Vehicle:** Elevate the vehicle using an appropriate lifting device, allowing the rear axle to hang freely. Secure the chassis with stands before proceeding.
- 2. Remove OEM Components:** Detach the existing original equipment manufacturer (OEM) bump stop and its associated bushing from the vehicle frame.
- 3. Assemble Helper Spring:** Combine the hollow rubber spring with the mounting bracket.
- 4. Mount Assembly:** Secure the assembled helper spring and bracket to the vehicle frame utilizing the existing mounting holes and the provided fasteners.
- 5. Lower Vehicle:** Carefully lower the vehicle back to the ground.

Installation Notes:

- The included 1" rubber spacer (Ref 5) is optional. Use it only when needed to reduce excessive unloaded clearance. Use the longer 3/8" x 2-3/4" hex cap screw (Ref 8) with the spacer and the shorter 3/8" x 1-3/4" screw (Ref 7) without it.
- Verify that no brake lines or cables interfere with the helper spring's position or movement.
- The Nylock locknut (Ref 9) may be used with offset holes on the bracket to precisely center the helper spring over the axle.
- If the OEM vehicle frame features a peg-style casting, the helper spring is designed to make contact with this peg. Do not remove the OEM peg. The bracket may need to be rotated for proper alignment with such a peg.

3. Clearance Instructions & Fitting Guidance

- **Clearance Gap:** With the vehicle unloaded and at normal ride height, maintain approximately **1/2" to 1-1/2" (13-38 mm)** of clearance between the rubber spring and its contact point. Approximately **1" (25 mm)** is a practical target.
- **Ride Comfort:** Proper clearance helps maintain a comfortable unloaded ride. Insufficient clearance or excessive preload may result in a firmer ride.

- **Spacer Use:** If the unloaded gap is too large, install the included **1" spacer** to reduce it. If clearance is already within the recommended range, omit the spacer. If the spacer causes little or no clearance, remove it.
- **Load Effects:** A partial load, cargo, or aftermarket vehicle accessories may reduce or eliminate this initial clearance.

Important Safety Warnings:

- Do not exceed the vehicle manufacturer's recommended axle weight rating / GAWR.
- Inspect for potential interference between the hollow rubber spring and adjacent vehicle components. Under load, the rubber spring will expand in diameter.

Note: Illustrations are for reference only. Vehicle configuration and actual component appearance may vary. Install according to the vehicle-specific mounting points and standard mechanical practices.